

GenderGap Analyzer: Data Analysis of Gender Wage Inequality

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Introduction

Gender wage inequality is a persistent global issue that affects economic growth and workplace fairness. Despite various policies and initiatives aimed at promoting equality, disparities in salaries and employment opportunities between male and female employees continue to exist.

With the advancement of data science, large datasets can be analyzed to uncover hidden patterns and trends in wage distribution. This project leverages data analysis techniques to study gender pay gaps across organizations. By examining salary differences, bonus distributions, and employee representation across pay levels, the study aims to provide insights into workplace inequality. The findings from this analysis can help organizations understand the extent of wage disparities and encourage data-driven decision-making for fair compensation practices.

Problem Statement:

The objective of this project is to analyze gender pay gap data across multiple organizations to identify patterns of wage inequality between male and female employees. The project focuses on examining hourly pay differences, bonus distributions, and employee representation across salary quartiles.

By applying exploratory data analysis techniques, the study aims to understand how gender-based disparities exist in compensation and workforce distribution. The results of this analysis can provide insights into the extent of inequality and highlight the need for improved policies to ensure fair pay practices.

Dataset Description

Dataset: Gender Pay Gap Dataset (paygap.csv)

- Total Records: ~48,711
- Attributes: 27 features

Key Features:

- diff_mean_hourly_percent
- diff_median_hourly_percent
- male_bonus_percent
- female_bonus_percent
- salary quartile distribution

Methodology:

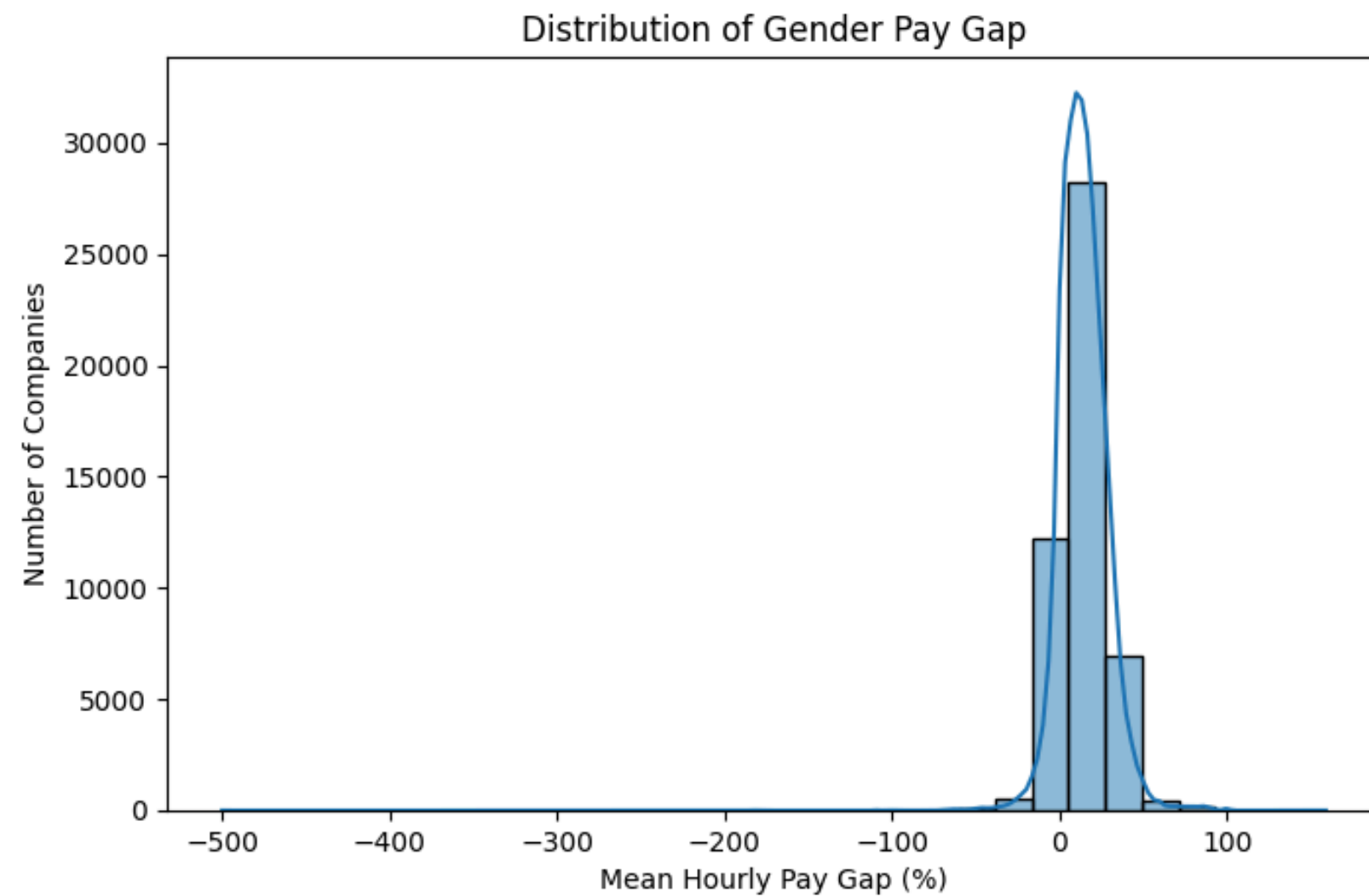
1. Data Collection
2. Data Preprocessing
3. Exploratory Data Analysis (EDA)
4. Data Visualization
5. Result Interpretation

Tools Used:

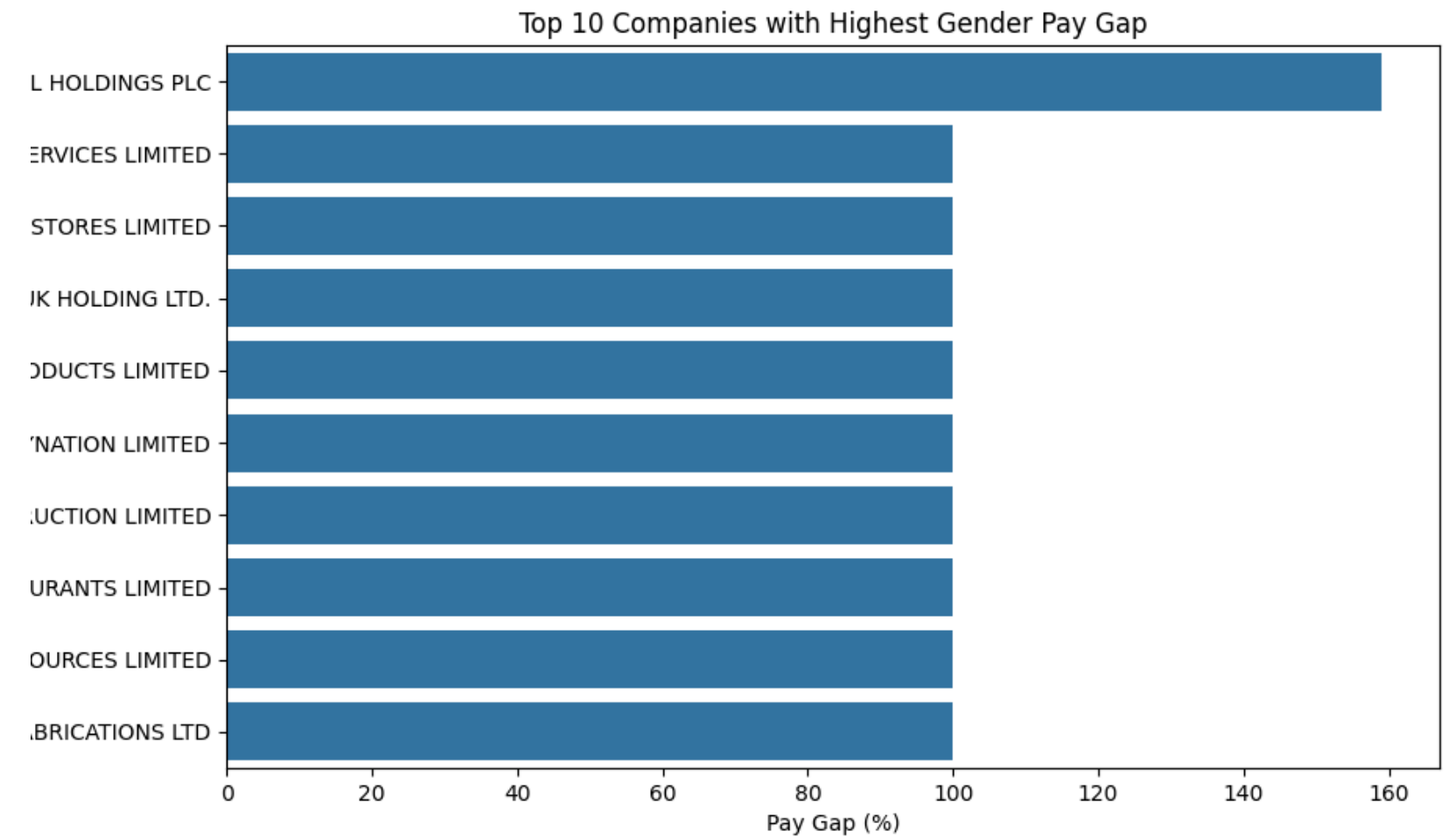
- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook

Visualizations

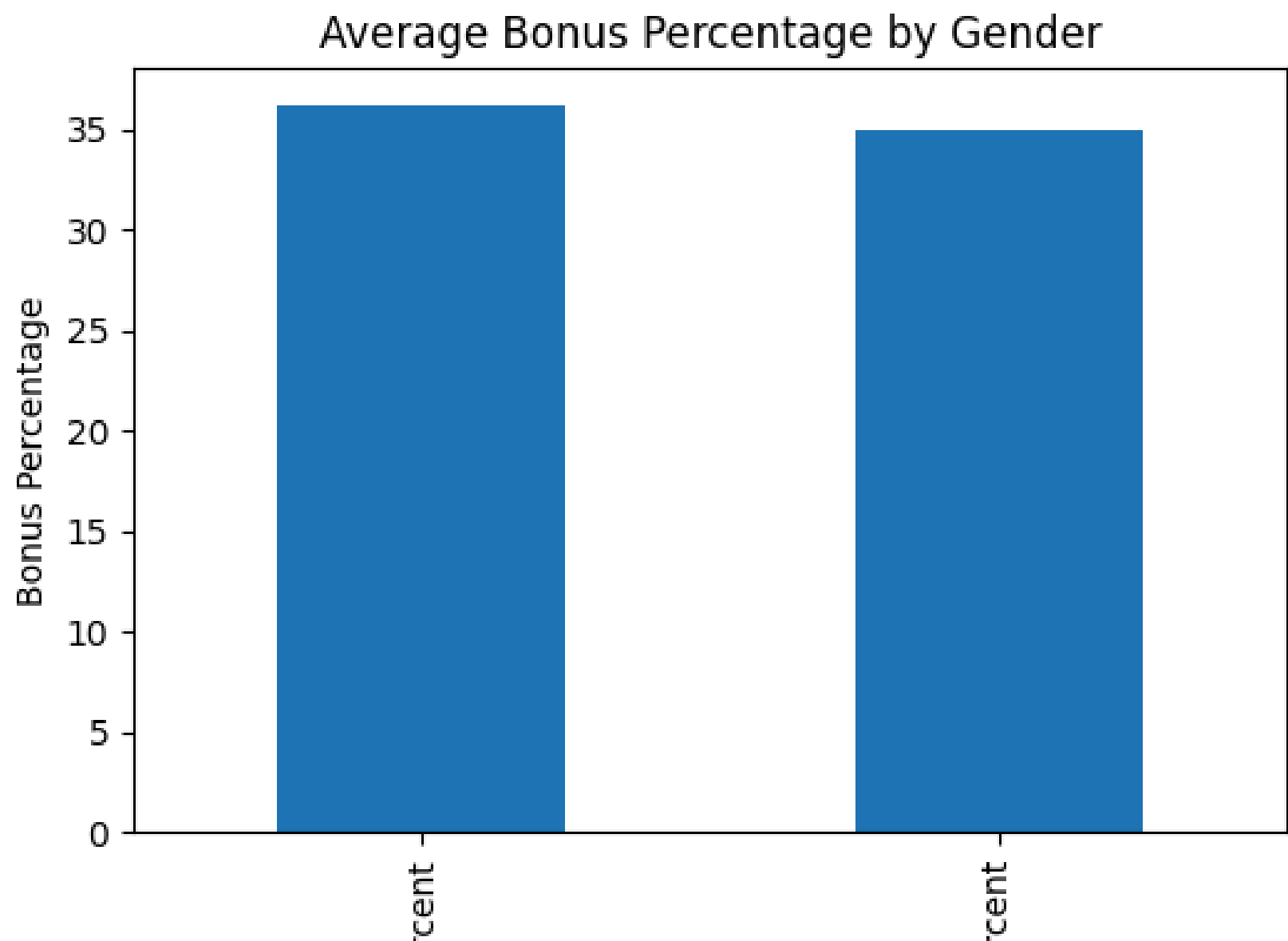
- Pay Gap Distribution (Histogram)



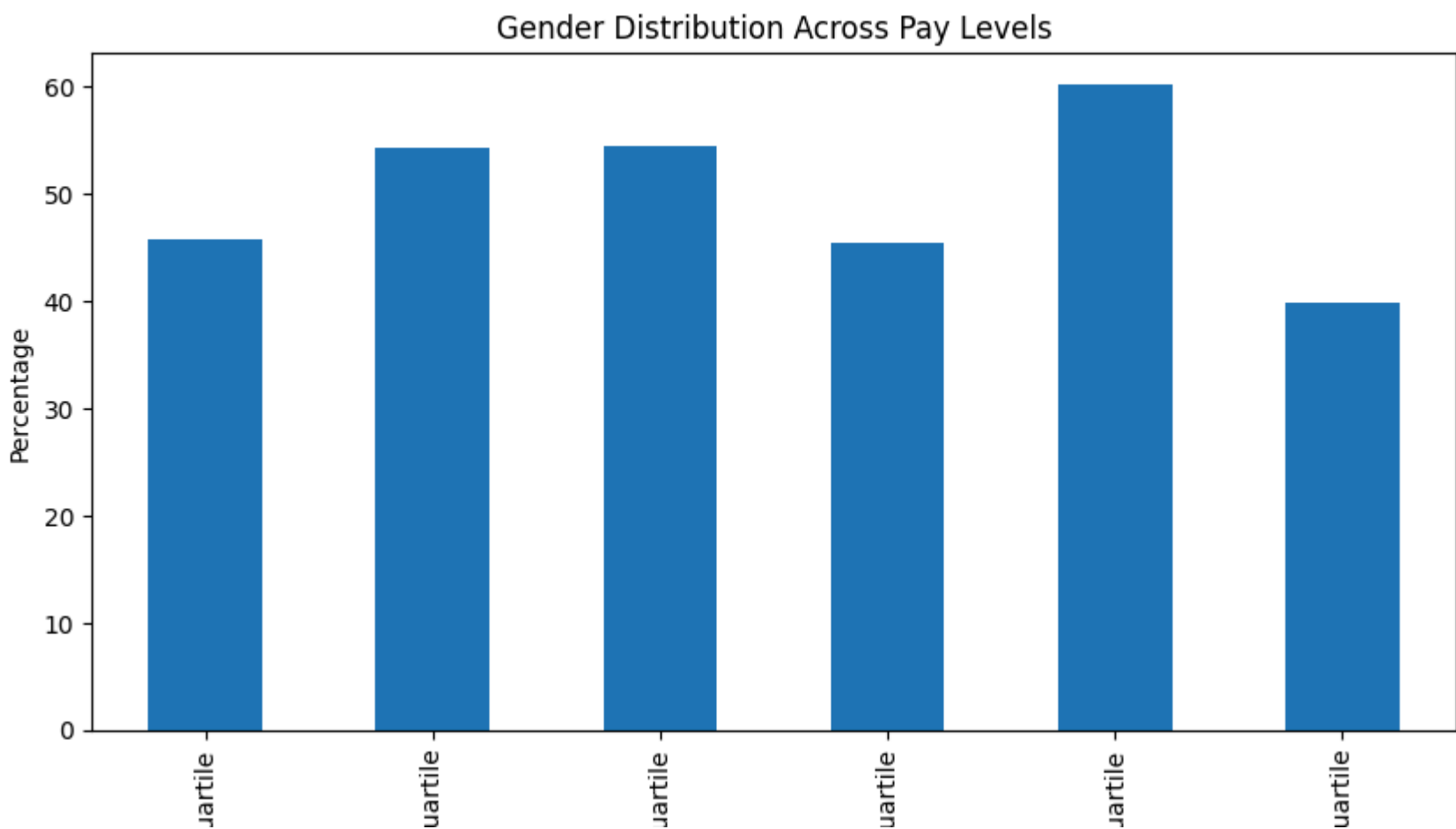
- Top Companies with Highest Pay Gap



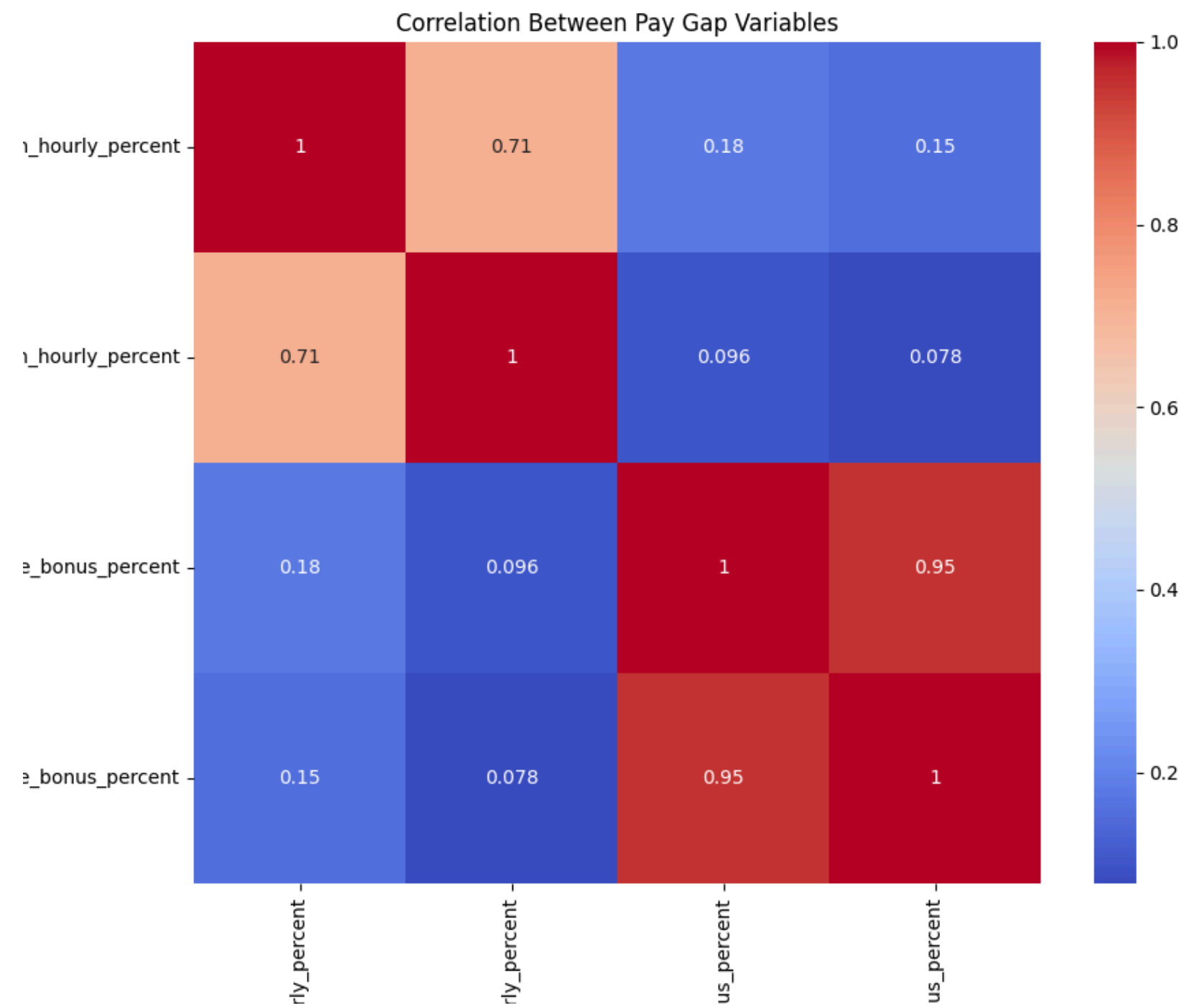
• **Bonus Comparison (Male vs Female)**



• **Gender Distribution Across Quartiles**



- Correlation Heatmap



Key Findings

- Average gender pay gap $\approx 14\%$
- Most companies show higher pay for male employees
- Male employees dominate top salary quartiles
- Female employees are more in lower salary quartiles
- Bonus differences are relatively small

Conclusion

- Gender wage inequality still exists in many organizations
- Data analysis helps identify salary disparities
- Male employees are more represented in higher pay levels
- Organizations should focus on fair pay policies
- Data-driven insights can support equality

GitHub

Project Repository:

https://github.com/sherlyisrael13/MiniProject_DS_AIML-A_2026_GenderGapAnalyzer

- Contains dataset, code, visualizations, and report
- Fully reproducible project

THANK YOU